CHAPTER

17

Complex Restorations

The restoration of extensively damaged tooth is always challenging. Many a times, the damage involves half or more than half of the tooth structure. The remaining tooth structure is not sufficient to retain the restoration and require modification of the classic preparation design. The earlier authors defined complex restorations as the restorations involving more than two tooth surfaces. The lesion on two or more than two surfaces, if small, the restoration can not be designated as 'complex'. The revised and accepted definition is *lesions, which require extra retentive device for their retention, are labeled as 'complex restorations'*.

The destruction of tooth structure, vis-à-vis, the remaining tooth structure is the decisive factor.

Location, as well as extent of coronal destruction, plays an important role in selection of restorative material and design of the cavity preparation. Location of destruction is categorized as *'peripheral'* (axial surfaces of the teeth, Fig. 17.1); *'central'* (central portion of the tooth, Fig. 17.2); and *'combined'* (incorporates both the surfaces, Fig. 17.3).

Minimal destruction of tooth structure in central, peripheral and combined locations can be restored with an amalgam or composite resin restoration. An amalgam or composite resin restoration can also be employed on moderately damaged teeth. However, combined destruction may require onlay, if the potential for fracture of unsupported cusp is evident (Table 17.1).

Severe destruction of the crown of posterior teeth can be treated by placement of pin-retained amalgam or composite resin core; followed by restoration of veneer crown.



Fig. 17.1: Peripheral tooth destruction

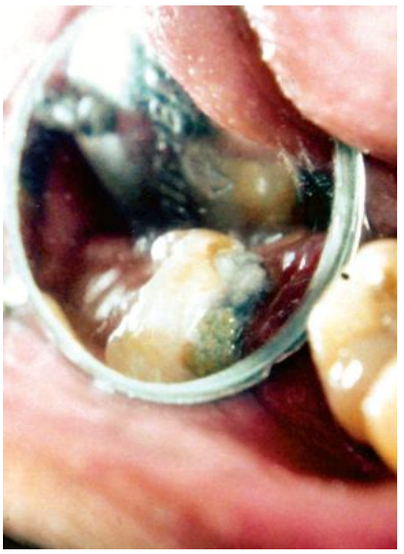


Fig. 17.2: Central tooth destruction

EXTRA RETENTIVE DEVICES

The extent of tooth structure lost is thoroughly evaluated. To compensate the lost tooth structure (may be moderate, severe or total), one or more than one retentive device is utilized (Table 17.2). Following features should be taken care of before placement of retentive devices:

- Central core (pulp and 1.0 mm thick surrounding layer of dentin) must not be invaded in vital teeth:



Fig. 17.3: Combined tooth destruction

Table 17.1: Categories of complex restorations: Treatment options

	Minimal	*Moderate*	*Severe*
Central	• Amalgam • Composite	• Bonded amalgam • Sandwich technique	Amalgam core + crown
Peripheral	• Amalgam • Composite	• Pin amalgam • Pin composite • Cast restoration • Crown	Full crown
Combined	• Amalgam	• Pin amalgam • Onlay	• Amalgam core + crown • Cast pin restoration

The retentive feature should be within 1.5 mm into the dentin from the cervical line and/or central fossa.

- No dentin wall be reduced to thickness less than 1.0 mm; minimum requirement for the foundation restorations.

The retentive devices used are:

i. ***Slots:*** Slots are horizontal grooves, created to increase the surface area for the restoration. A 0.75–1.0 mm deep and 1.0– 3.0 mm wide cut is created on the pulpal and/or cervical wall. This rectangular depression in these walls increases the bulk for the restoration, and also increases the area for bonding (Fig. 17.4A).

ii. ***Grooves/locks:*** The locks are given in direct restorations while the grooves are given in indirect restorations. The major difference between the groove and the lock is that the lock extends up to the pulpal wall (from cervical to pulpal) and the groove extends up to the cavosurface margin (from cervical to cavosurface). The grooves maintain the parallelism; while, the locks converge occlusally. The grooves/locks are given at axiobuccal and axiolingual line angles, inside dentinoenamel junction (Fig. 17.4B).

iii. ***Crown lengthening:*** The compensating procedures to lengthen the crown surgically are followed in routine (Fig. 17.5). The exaggerated apical placement of finish line may result in apical migration of epithelial attachment and alveolar crest of bone. (*Alveolar crest may migrate to a level of approximately 3.0 mm apical to the margin of the restoration.*) The crown-root ratio may also be compromised.

iv. ***Pins:*** Pins have long been utilized for retention purposes in restorative dentistry.

A complete knowledge about characteristics of retentive devices is desirable for better results.

Table 17.2: Compensation for lost tooth structure

Loss of tooth structure	*Features*	*Compensating procedures*	*Final restoration*
Moderate	One or two cusps missing, but more than 50% of coronal tooth structure remaining. Thickness of remaining walls: at least 1.0 mm	• The preparation should include: – Grooves – Pin	Full/half veneer crown
Severe	Loss of 50% or more of crown. Thin walls (less than 1.0 mm)	• Core with pins • Lengthen the crown surgically, if need be	Full veneer crown
Total	All cusps destroyed; supragingival height 1.0 mm or even less	• Place amalgam or composite resin core with pins • Lengthen crown surgically, if need be	Full veneer crown

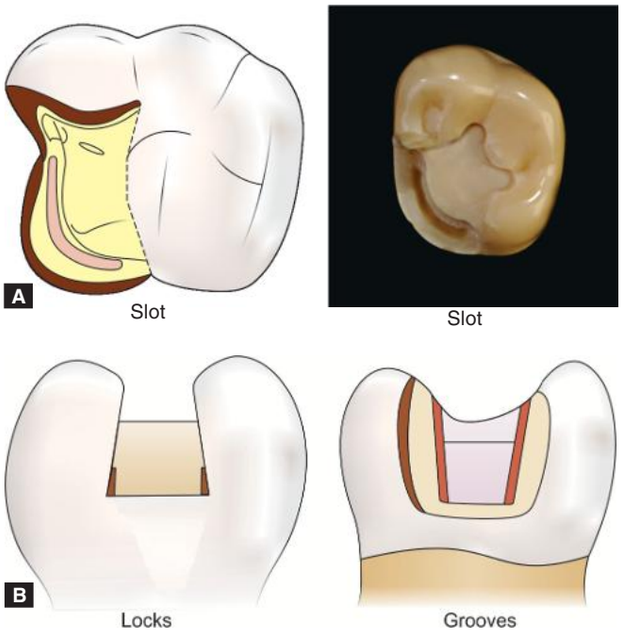


Fig. 17.4: Extra retentive devices: **(A)** Slot; **(B)** Locks/grooves



Fig. 17.5: Crown lengthening

PIN-RETAINED RESTORATIONS

Conventional cavity preparation in badly mutilated cases requires removal of large amounts of tooth structure for obtaining retention and resistance forms. Pins provide efficient and adequate retention to the restorations with the least sacrifice of healthy tooth structure.

It is emphasized that pins are auxiliary aids of retention; should be used only if primary retentive features are not sufficient to provide the requisite retention.

Indications

- Badly broken down/mutilated teeth; large cavity preparations.
- Used in foundations for full veneer restorations.
- Pin-retained cores facilitate rubber dam application; subsequently, facilitating restorative procedures.

Advantages

- Provide retention without the need for extensive preparation of tooth structure.
- Increase resistance form of the preparation (some extent).
- Relatively less time consuming than cast restorations.
- Comparatively cost-effective.

Disadvantages

- Do not increase the compressive strength of the overlying restorative material; may reduce the transverse and tensile strengths.
- Induce stresses in dentin, which may increase the potential for fracture of tooth.
- Increase the chances of perforations into pulp or on external tooth surface.
- Chances of microleakage exist at the pin–dentin interface, if the overlying restoration leaks.

Classification

Pins can be classified as:

1. *Direct pins:* Direct pins, also known as non-parallel pins, are usually made of stainless steel, silver, titanium, etc. These pins are inserted directly into the tooth structure; hence, need not be parallel.
2. *Indirect pins:* Indirect pins, also known as parallel pins, are an integral part of a cast restoration. The pins are held in the pinholes by means of a cement. The pinholes are created in the dentin and the impression of pinhole along with the rest of preparation is cast (pin and cast in one unit).

1. *Direct Pins*

The direct pins are of three types:

a. ***Cemented pins:*** Cemented pins are 0.0013"–0.0023" smaller than their pin channels and the difference in diameter provides space for the cementing medium. Cemented pins are generally indicated in cases where least crazing and stresses are desired in the remaining tooth structure, e.g. endodontically treated teeth. It is the preferred technique for Class IV preparations, where the pins need to be bent.

Technique: The method utilizes pins in the form of stainless steel wires, which are threaded to the required length using wire cutter or Dial-A-Pin cutter. Pinholes extending 2.0–4.0 mm into dentin are prepared using twist drills. The length should be such that after seating pin into the pin channel completely, the pin extends not more than 2.0–3.0 mm above the base. If necessary, the pins can be bent with contouring pliers to keep them within the contours of the restoration. Any roughness on the protruding end of the pin is smoothened.

Advantages

- Cemented pins are approximately 0.001"–0.002" (0.025–0.050 mm) smaller than their pinholes and hence comfortably seated.
- Create no stress on the surrounding dentin during or after placement.
- Minimal microleakage at the pin–tooth interphase.
- Pin can be bent as desired.

Disadvantages

- Less retentive as compared to the friction locked and threaded pins.
- Difficult to insert cement into the pinhole.
- Poorly cemented pin may get dislodged when the filling material is being inserted.
- Procedure is time consuming.

In teeth where more than one pin is to be inserted, each pin is selected and kept aside, until cementation. The pins are so kept that they allow easy identification of the pinholes for which they are intended.

The pin channels are dried with paper points. For cementation purposes, zinc-phosphate cement, polycarboxylates and light cure glass-ionomer cements are commonly used. The cement mix is introduced into the pin channel with a root canal file. The pin held in the forceps is also coated with the cement and inserted into the hole. The pin is held in its position till it sets. The excess cement is flicked off with an explorer point.

b. *Friction-locked pins:* The diameter of friction-locked pin is 0.001" larger than the pin channel; utilize elasticity of dentin for retaining the pins (pin of size 0.022" is inserted into a pinhole of size 0.021"). These pins are indicated in teeth that are vital and periodontally sound, and where direct access is possible so that the tapping force can be applied parallel to the long axis of the pin. These pins are preferred when sufficient amount of dentin is available to surround the pin.

Advantages

- Better stability.
- Better retention than the cemented pins.
- Cement is not required; no cement related problems.

Disadvantages

- Length of the pin is judged by trial; cannot be removed from dentin, once inserted.
- Bending/contouring of the pin after insertion may lead to further stresses.
- Driving pins into their respective pinholes generates stresses in dentin.
- In case the pin does not reach the full depth of the channel, the retentive capacity is compromised.
- Microleakage is higher as compared to cemented pins.

Technique: A spiral drill mounted in a low-speed handpiece is used to prepare the pin channel in dentin to a depth of 2.0–3.0 mm (1.5 mm inside the dentino-enamel junction). It is advisable to select the desired length extraorally, as the pin cannot be inserted into the pinhole for try-in and adjustment. The depth to which the pin channel has been drilled is marked on the pin with a permanent marker. The pin is inserted into a pin setter and carried to the pinhole. A mallet is used to apply force parallel to the axis of the pin. Do not use cement under any circumstances.

c. *Threaded pins:* The diameter of threaded pins is 0.0015"–0.002" larger than the pin channel, like the friction locked pins they are also retained by the elasticity of dentin. Additionally, they actively engage the tooth structure through their threads *(similar to a screw inserted into a wooden block)*. These pins are indicated in vital teeth and where maximum retention is desired. These pins should be given only when sufficient amount of dentin is available.

Four varieties of threaded pins are available depending on their sizes (in decreasing order of sizes) as:

- Regular : 0.030", 0.031"
- Minim : 0.024", 0.025"
- Minikin : 0.019", 0.020"
- Minuta : 0.014", 0.015"

i. The standard pin is a full length pin, i.e. 7.0 mm long, which can be cut to the required length after placement.

ii. The self-shearing single pin is designed so that when it reaches the bottom of the pinhole, the head separates automatically at the shear line, leaving a portion, which project from the dentin. A flattened head on one end of the pin is shaped to engage the slot in the hand wrench or the handpiece chuck (Fig. 17.6A, B).

iii. The two-in-one design is one in which two pins are connected to each other at a joint. This joint serves as a shear line for the peripheral pin. The two-in-one pin is approximately 8.0–9.0 mm in length and provides two pins of equal lengths.

iv. The disposable latch head design has a plastic sleeve/head designed to fit in a slow speed contrangle handpiece. The pin appears to float freely in the plastic sleeve. This aids in self-alignment as the pin is driven into the pinhole.

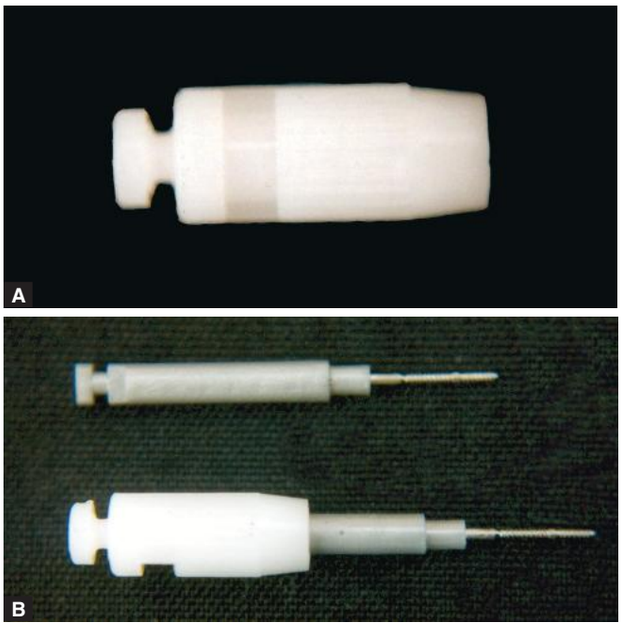


Fig. 17.6: Threaded pins: **(A)** Hand wrench; **(B)** Pin and pin in the hand wrench

After the pin reaches the bottom of the pinhole, the resistance offered causes the head to separate from the pin at the shear line.

A slight variation to the above design is the incorporation of sharper threads, a shoulder stop and a tapered tip, known as the 'Link Plus Series' (Fig. 17.7).



Fig. 17.7: Modified threaded pins (two types): Link series (red), Link plus series (white)

Advantages
• Ease of insertion. • Maximum retention.

Disadvantages
• Excessive stresses are generated in the dentin. • Contouring/bending of pin may create extra stress on the tooth. • When the pin is forced into the channel, it may strip the sides of the dentin. • Complete seating is difficult. • Microleakage is higher as compared to cemented pins.

Technique: Prepare the pin channel in dentin to a depth as instructed by the manufacturer depending upon the length of the pin. Pins are inserted either manually or mechanically.

For manual insertion, pins are attached to the hand wrench and threaded slowly into position. Tactile sense is a major factor in determining whether the pin has reached the bottom of the pin channel. For mechanical insertion, pins are engaged in the handpiece chuck. With the handpiece operating at slow speed, the pin is inserted with light pressure. On reaching the full depth, resistance is increased and the pin shears off at the shear line or disengages from the handpiece drive.

'L' and 'T' shape pins are also available to avoid bending.

PRINCIPLES OF PIN PLACEMENT

The principles involved in placement of pins are:

1. ***Number of pins:*** One pin per missing cusp or one pin per missing axial line angle (Figs 17.8 and 17.9) is the thumb rule to decide the number of pins. Increasing the number of pins enhances the chances



Fig. 17.8: Pin channel preparation



Fig. 17.9: Multiple pin channels prepared

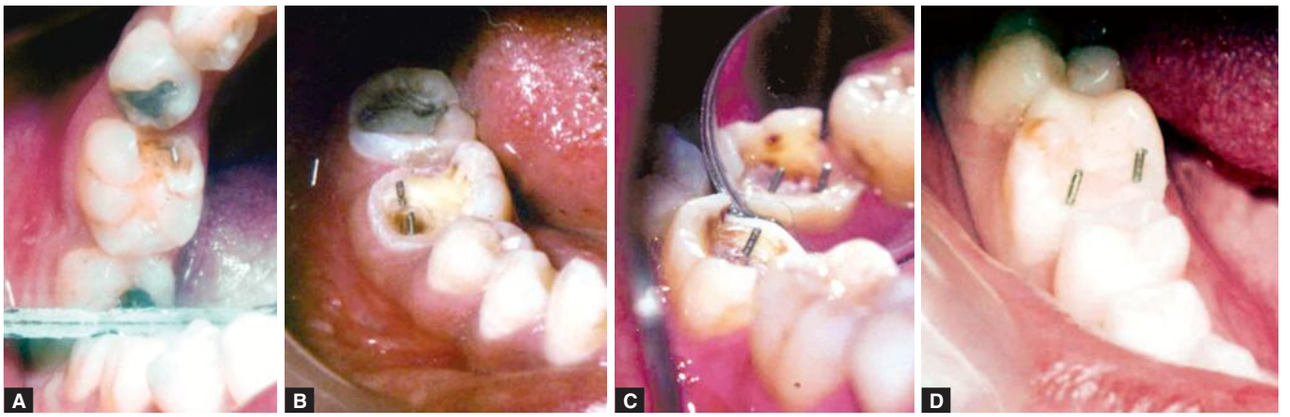


Fig. 17.10: Location of pin placement: **(A)** At the mesiolingual line angle of the tooth; **(B)** At the mesiobuccal and distobuccal line angles of the tooth; **(C)** At the mesiolingual and distolingual line angles of the tooth; **(D)** At the mesiobuccal and mesiolingual line angles of the tooth

of increased stresses (*aim is to achieve adequate retention with minimum number of pins*).

2. ***Site of pin:*** Adequate knowledge about the pulpal anatomy and external contours of the concerned tooth is important to decide about site of the pin placement.

   Patient's age is also important as young patients have higher pulp horns. The most desirable locations for pinholes are the facio-/linguo-proximal line angles (Fig. 17.10A to D) or corners of the tooth. The least desirable locations are in the middle of the facial, lingual, mesial and distal surfaces of a tooth. The operator should place pins in locations, which are surrounded by optimum bulk of dentin.

   The features affecting location for pinholes are as follows:

   - The site should be half way between the pulp and the dentinoenamel junction/external surface of the tooth.
   - There should be at least 1.0 mm of sound dentin around the circumference of the pin.
   - Pins should be located at least 1.0 mm inside the dentinoenamel junction to avoid crazing.
   - Dentin thickness around each pin should be at least 2.5 times the diameter of the pinhole.
   - Pin placement should allow at least 1.0 mm of dentin thickness between the pulp and the pin.

   A cove is prepared using appropriate bur, which creates space for pinhole preparation as well as allows a clearance of 0.5 mm around the pin for proper condensation of material (Figs 17.11 and 17.12A to C).

   Pinholes should be located on flat surfaces. It is difficult to stabilize a pin drill on an inclined surface and moreover an inclined plane interferes with the shoulder stop of a drill preventing its penetration to full depth.



Fig. 17.11: Cavity preparation in which pin placement would face problem (pin placement would be close to the longitudinal wall of the cavity)

3. ***Pin orientation:*** Pins should be oriented parallel to the long axis of the tooth. However, in excessively cervically placed areas towards the cemento-enamel junction, it is mandatory to direct the pin parallel to the closest external surface of the tooth and then bend it slightly.
4. ***Pin diameter:*** The selection for the diameter of the pin depends upon: (a) the amount of dentin available, (b) the size of the concerned tooth, and (c) the amount of retention required. Generally, increasing the diameter of the pin offers increased retention, but large-sized pins are also associated with a heavy concentration of stresses in dentin. Preferably, the depth should be more than the diameter.
5. ***Pin length:*** The optimum ratio of pin length in dentin to pin length in restorative material varies with the different types of pins, i.e. for cemented pins, 3.0 mm and 2.0 mm; for friction locked pins, 3.0 mm and 3.0 mm; and for threaded pins, 2.0 mm and 2.0 mm, respectively. Preferably, the length of

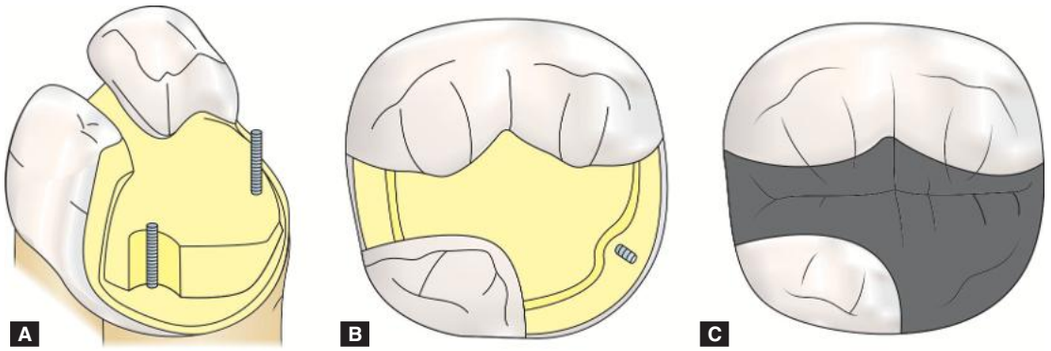


Fig. 17.12: Pin placement: **(A)** In case pin placement is close to the longitudinal wall of the cavity, a cove is prepared in the adjacent vertical wall to create space for pin placement (proximo-occlusal view); **(B)** Mesio-occluso-distal cavity in mandibular second molar with one pin replacing the distolingual cusp (occlusal view); **(C)** Completed pin-retained restoration (diagrammatic)

the pin inside the dentin and the length of the pin in restorative material should be equal; or length of pin in dentin should be more than the length of pin in the restorative material.

6. ***Interpin distance:*** The optimum interpin distance depends upon the type and size of pins. The minimal interpin distance is 2.0 mm for cemented pins, 4.0 mm for friction locked pins, 3.0 mm for Minikin threaded pins and 5.0 mm for Minim threaded pins. The accepted principle is to have 2.0 mm of dentin around each pin.

PINHOLE PREPARATION

Pinhole is prepared using a device called 'twist drill' (Fig. 17.13); an instrument that performs cutting when rotated clockwise at low speeds. The recommended speed for drilling channels is 300–500 rpm (ultralow speed) or 1000 rpm (low speed). Very little heat is generated at these speeds; hence, water or air cooling is not required.

The proposed pinhole location is marked with a marker before the pin channel is prepared with the twist drill. A No.1/4 round bur at low speed is dipped to about half the diameter at these marks to create starting points, known as lead holes or pilot holes. The purpose of these holes is to permit more accurate positioning of the twist drill and to prevent its skidding out laterally into unfavorable locations.

Slight pressure is applied in an apical direction when the drill starts cutting into dentin until it penetrates to its marked length. The drill should be dipped only once in dentin and quickly removed allowing only a brief period of rotation at the bottom of the pinhole. It should be rotating when it enters and leaves the dentin.

Stepwise procedure for pin-retained amalgam restoration is shown in Fig. 17.14A to E. Similarly, pin-retained composite restoration is shown in Fig. 17.15A to C.



Fig. 17.13: Twist drill for making pinholes

Pin placement: Key points

- Repeated insertion and removal of drill in the same channel can result in enlarged channel.
- Lateral stresses applied while drilling can induce additional stresses in dentin; increase the chances of drill breakage.
- Rotating pin drills for longer period after they have reached the bottom of the pin channel can result in large-sized holes.
- Stopping the pin drill during the drilling procedure increases the chances of drill breakage.

PINS, STRESSES AND TOOTH

Stresses are always associated with insertion of friction locked and threaded pins in dentin. Insertion of larger pins into smaller pin channels is possible because of elastic properties of dentin. Once the elastic limit of dentin is exceeded, plastic deformation occurs. If the stresses sum up to a point where the dentin's elastic limit is exceeded, microscopic craze lines are generated.

Friction-locked pins act as wedges and lateral stresses are the main type of stresses present around them, compared to threaded pins, which are associated mainly with apical stresses.

The factors, which increase the stresses in dentin, are:

- Pin morphology:
  - Large diameter of the pin
  - Large difference between the pin and pin channel
  - Narrow-pitched threads (greater number of threads per unit distance)
  - Excessively long threads

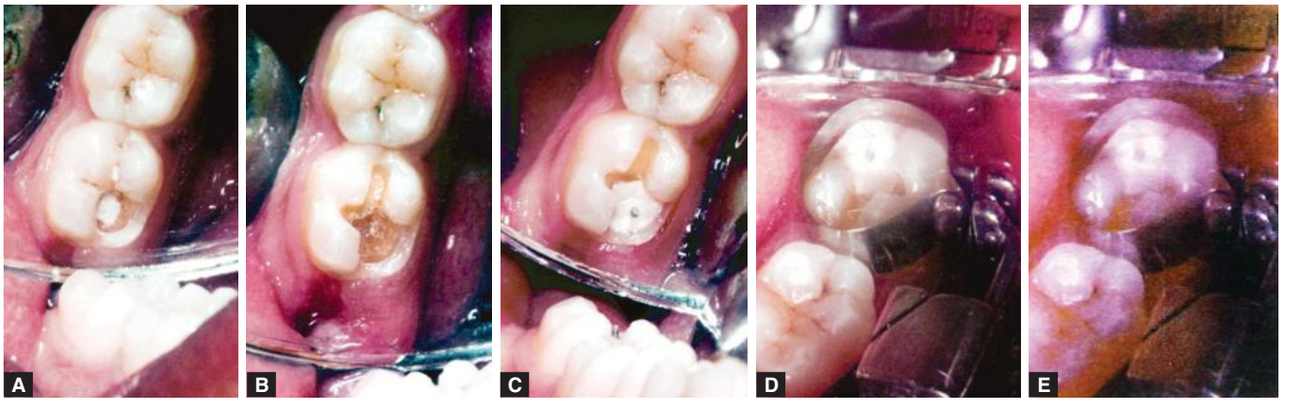


Fig. 17.14: Pin-retained amalgam restoration: **(A)** Preoperative (distolingual cusp of mandibular right second molar is lost); **(B)** Cavity preparation; **(C)** Glass-ionomer cement base applied and pin inserted on the distolingual line angle of the tooth; **(D)** Tofflemire retainer and band applied; **(E)** Final restoration (finished and polished)

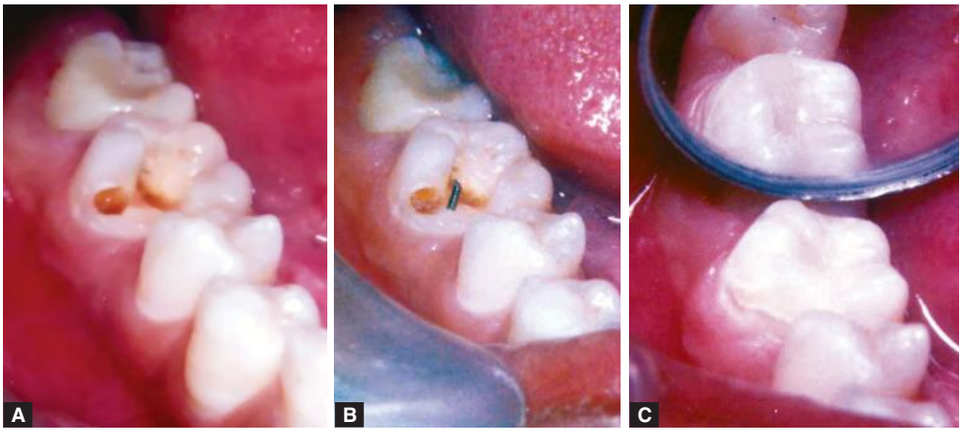


Fig. 17.15: Pin-retained composite restoration: **(A)** Carious lesion with mesiobuccal cusp lost on mandibular right first molar; **(B)** Cavity preparation with pin placed at mesiobuccal line angle; **(C)** Completed pin-retained composite restoration

  - Irregular pin ends
  - Inserting a flat-ended pin into a chisel-ended channel
- Placing pins close to each other causes interaction of the field of stresses of individual pins.
- Increasing the number of pins per tooth increases stresses.
- An attempt to drive the pin further into its channel increases stresses.
- Bending may induce stresses.
- Cutting channels with dull drills increase stresses.
- Lesser the bulk of dentin surrounding the pin, greater are the stresses per unit volume of dentin.

Pins are likely to reduce the strength of amalgam and composite restorations, because of the absence of any chemical union between pin and the restorative material at the interface. The pin is believed to serve as a void in restorative material; hence, act as a stress initiator.

The factors, which decrease the compressive and tensile strength of a restoration are:

- Pin in restoration may be wedge-shaped; serving as areas of stress concentration.
- Pins are close to the outer surface of the restoration. This decreases the bulk of material between pin end and restoration surface, thereby not providing adequate strength.
- Pins are closer than 2.0 mm to each other (increases the possibility of incorporation of voids).

FACTORS AFFECTING RETENTION OF PINS

I. Tooth Structure

The factors affecting retention of pins in tooth structure are:

a. ***Pin type:*** Among the three systems of pins, cemented pins are the least retentive. Friction-locked pins show intermediate retention and threaded pins are the most retentive. Friction-

locked and threaded pins are retained by the elasticity of dentin and this accounts for their higher retention. Within the cemented pin technique, serrated or threaded pins offer more retention than smooth surface pins.

b. ***Pin depth in dentin:*** Increasing the depth of pin in dentin increases the retention. The relationship is linear for the cemented pins; however, non-linear for the friction-locked and threaded pins (increase in depth do not increase retention).

c. ***Pin diameter:*** As the diameter of the pin increases, the retention increases. However, the practical limitation of having appropriate amount of dentin around pin would warrant use of smaller diameter pins.

d. ***Pin number:*** Increasing the number of pins increases the retention; however, increase in number of pins increases crazing of dentin.

e. ***Cementing agents:*** Zinc phosphate cement is more retentive than polycarboxylate cement; better than zinc oxide eugenol cement. Glass-ionomer cement is also adequately retentive.

f. ***Association between pin channel and pin circumference:*** The pin channel circumference should closely match the pin circumference. A pin with circular cross-section will show less retention when placed in elliptical pin channels.

g. ***Miscellaneous:*** The primary dentin provides better retention than secondary dentin.

II. Restorative Material

The factors affecting the retention of pins in restorative material are:

a. ***Pin length:*** Increasing the length of the pin in restorative material increases retention. The relationship is linear for friction-locked pins but non-linear for cemented and threaded pins. (*2.0 mm into dentin and 2.0 mm into the restorative material is considered ideal.*)

b. ***Pin number, diameter and orientation:*** Increasing the number of pins poses problem in condensation of restorative material and also decreases the overall strength. Retention increases with the increase in diameter of the pin. Pin orientation affects retention, i.e. pins placed in a non-parallel manner increase retention.

c. ***Surface characteristics of the pin:*** The number of surface deformations per inch influences the retention of pins in restorative material.

d. ***Pin shape:*** The square- or pear-shaped pinheads improve retention.

e. ***Interpin distance:*** Interpin distance less than 2.0 mm leads to reduction in pin retention, because the less amount of material present in between the pins.

f. ***Type of restorative material:*** Restorative material with a greater wetting ability provides closer adaptation to the pin surface, and hence greater frictional retention. Spherical amalgam alloys exhibit superior adaptation to all three types of pins; whereas, lathecut alloys exhibit good adaptation to only friction-locked pins.

g. ***Bulk of material surrounding the pin:*** Pin retention decreases with the decrease in bulk of material surrounding the pin.

COMPLICATIONS DURING PIN PLACEMENT

1. ***Drill breakage:*** The drill may break in following circumstances:
   - Lateral stresses during drilling
   - Use of dull drills
   - Drill is allowed to stop rotating while entering or exiting from dentin.
2. ***Pin breakage:*** The factors causing pin breakage are:
   - Excessive force during driving the pin into its pin channel
   - Pin is rotated after seated in the channel
   - Careless bending the pin.
3. ***Loose pin:*** The pin can be loose in the channel, because:
   - Repeated insertion and removal enlarge the pin channel
   - Poor quality control between pin drill and the pin sizes
   - A self-shearing pin fails to shear; resulting in stripped out dentin.

   (A well-fitted pin in the channel may also get loosened while shortening at a later stage).
4. ***Twist drill dulling:*** Cutting edges of a twist drill are dulled when the drill is repeatedly used. Dull drills have decreased cutting effectiveness, induce more stresses in dentin, and are more prone to breakage.
5. The pin fails to bind and shear, but keeps rotating within its channel.
6. ***Heat generation:*** Rise in temperature, subsequent to the use of twist drills, may be related to cutting deep channels, high rotational speed used during channel preparations, large-sized twist drills and repeated insertion of pin drills into dentin.
7. ***Microleakage:*** Microleakage in dentin is evident under all types of pins and is related to the leakage potential of different restorative materials. With the use of amalgam, leakage generally diminishes in time due to accumulation of corrosion products. On the other hand, resin restorations show a gradual deterioration in the margin seal as a function of time.

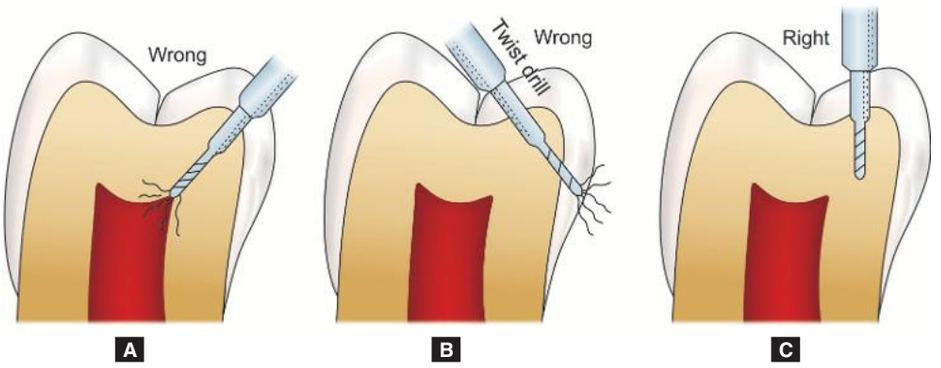


Fig. 17.16: Wrong angulation of the twist drill may lead to perforation; **(A)** In pulp chamber; **(B)** On external surface of tooth; **(C)** Correct angulation of a twist drill

8. ***Dentinal cracks:*** Microcracks usually form subsequent to introduction of friction-locked and threaded pins into dentin.
9. ***Perforation into the pulp space:*** The pin placement may perforate the pulp (Fig. 17.16A). The exposure will lead to:
   - Bleeding in the pinhole following removal of the drill from the pin channel.
   - Sudden lack of resistance is felt.
   - Sudden pain while drilling, if the tooth is not anesthetized.
   - The pin continues to thread greater than its intended depth.

   (Radiographs may provide some information).
10. ***Perforation onto the external tooth surface:*** The external tooth surface (periodontium) may get perforated (Fig. 17.16B) and lead to:
    - Bleeding in the pinhole following removal of pin drill.
    - Lack of resistance.
    - Pin continues to thread greater than its intended depth.
    - Pain is felt by the patient when the pin hole is being inserted.

Radiographs can verify an external perforation when the pin seems to project outside the tooth. Additionally, observing the direction of a pin drill or pin indicates a possible lateral perforation.

The incidence of pulpal and periodontal perforations can be greatly reduced, if the clinician is thorough with the pulpal and external anatomy of the tooth. Correct angulation of pin drill is shown in Fig. 17.16C.

EFFECT OF PINS ON PULP

Pins may be placed close to the pulp or there may be a minute pulp exposure during drilling.

Generally, the pulp responds positively to the pin and accepts its presence without any adverse effects.

The proximity of pin channel to the pulp tends to induce initial pulpal response, like destruction of odontoblastic nuclei, vascular dilation and inflammatory cell infiltrations. Inflammatory reactions have been observed under all types of pins. The response under cemented pins is attributed to the type of cement used and its effect on pulp. Zinc phosphate cement without an underlying varnish layer induces a more severe inflammatory response as compared to Zinc oxide eugenol or Glass-ionomer cement. Pulpal response under friction-locked and threaded pins is probably because of the microcracks generated in communication with the pulp or disruption of the odontoblastic processes. The pulpal response is succeeded by normal defense reaction, like formation of reparative dentin under all three pins.

AMALGAPINS

Shavell (l980) introduced the concept of amalgapins as an alternative to the traditional stainless steel pins. It is referred to as a vertical post of amalgam anchored in dentin (Fig. 17.17). Amalgapins offer almost similar retention when compared to dentin retained pins, yet seem to cause less internal stress in the dentin. They also do not reduce the compressive and transverse/tensile strengths of amalgam, unlike pins. However,

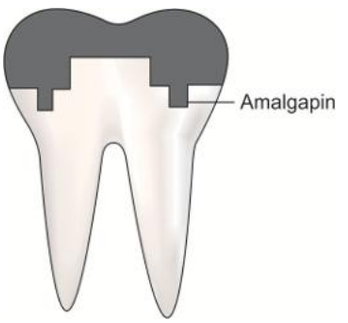


Fig. 17.17: Cavity preparation using amalgapins

they are associated with a greater tooth substance removal.

An inverted cone bur of appropriate size is used to create a ditch, 2.0 mm deep, 0.5–1.0 mm wide, and 1.5 mm inside the dentinoenamel junction. Amalgam is condensed directly into the channel with fine condensers.

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